1 Models

Robust NLME model for the longitudinal response

The one-compartment exponential decay NLME model:

$$\begin{aligned} y_{ij} &= P_{i1} + P_{i2}e^{-\lambda_i t_{ij}} + e_{ij}, \\ P_{i1} &= P_1 + u_{i1}, \quad P_{i2} = P_2 + u_{i2}, \quad \lambda_i = \lambda + u_{i3}, \end{aligned} \tag{1.1}$$

for $i = 1, \cdots, n$; $j = 1, \cdots, n_i$, where y_{ij} is the $\log_{10}$-transformation of the viral load measured at time t_{ij} for individual i, and the residual e_{ij} and the random effects $\boldsymbol{u}_i = (u_{i1}, u_{i2}, u_{i3})^\top$ are independent. We assume fixed variance components for the random effects $\boldsymbol{u}_i$, that is $\mathrm{Cov}(\boldsymbol{u}_i) = D$. Denote the diagonal elements of D by (d_1^2, d_2^2, d_3^2), thus d_1, d_2, d_3 are the standard deviations of random effects u_{i1}, u_{i2}, u_{i3}, respectively.

Model specification for the residual variance:

The residual e_{ij} are assumed to be i.i.d. $\sim \mathcal{N}(0, \sigma_{ij}^2)$, where the variance σ_{ij}^2 depends on individual i, the time t_{ij} and another covariate, CD4. Given x_{ij}^*, the "true" but unobservable (log-) CD4 cell count at time t_{ij} for individual i, we consider the following LME model for the residual variance:

$$\log(\sigma_{ij}^2) = \alpha_0 + \alpha_1 x_{ij}^* + a_i, \tag{1.2}$$

for $i = 1, \cdots, n$; $j = 1, \cdots, n_i$, (α_0, α_1) are fixed variance parameters and a_i is a random effect. We assume $a_i \sim \mathcal{N}(0, \sigma_a^2)$.

It is worth mentioning that model (1.2) links the residual variance to the unobserved true value of covariate x_{ij}^* rather than the observed value x_{ij}. In practice, the longitudinal clinic data, like CD4 cell counts, are often measured with measurement errors. Directly using the observed data with ignoring the measurement errors may lead to misleading statistical inference. Therefore, we consider another LME model to the covariate process to address the measurement errors..

LME model for the covariate process

Covariate model for CD4:

$$x_{ij} = x_{ij}^* + \varepsilon_{ij} \equiv \gamma_0 + \gamma_1 t_{ij} + \gamma_2 t_{ij}^2 + b_i + \varepsilon_{ij}, \tag{1.3}$$

where we assume that the unobserved true trajectory for CD4 is $x_{ij}^* = \gamma_0 + \gamma_1 t_{ij} + \gamma_2 t_{ij}^2 + b_i$, b_i is a random effect, and ε_{ij} stands for the measurement error. We assume $b_i \sim \mathcal{N}(0, \sigma_b^2)$, $\varepsilon_{ij} \sim_{iid} \mathcal{N}(0, \xi^2)$ and they are independent from each other.

2 Inference Methods

Two-step method

A straightforward approach is a Two-step inference method which fits the models to the observed data separately and substitutes the unobserved covariates by their estimates. The inference is made for each model separately as if the estimated variables were observed data. To be more explicit, the Two-step inference method for joint models (1.1), (1.2) and (1.3) can be conducted in the following two steps:

1. Fit the LME model (1.3) to the observed data $\{(x_{ij}, t_{ij}), i = 1, \cdots, n; j = 1, \cdots, n_i\}$ and get the estimated true covariate values $\{\tilde{x}^*_{ij}, i = 1, \cdots, n; j = 1, \cdots, n_i\}$;

2. Fit the proposed robust NLME model (1.1) (with variance model (1.2)) to the "observed data" $\{(y_{ij}, t_{ij}, \tilde{x}^*_{ij}), i = 1, \cdots, n; j = 1, \cdots, n_i\}$, where the unobserved covariate x^*_{ij} is substituted by its estimates $\tilde{x}^*_{ij}$.

The parameters in the LME model (1.3), $\{\boldsymbol{\gamma} = (\gamma_0, \gamma_1, \gamma_2)^T, \sigma_b, \xi\}$, are estimated in Step 1 by using the regular likelihood inference. While the parameters in the robust NLME model (1.1)+(1.2) are estimated separately in Step 2 by using the h-likelihood inference.

Let $\boldsymbol{\beta} = (P_1, P_2, \lambda)^T$ denote the fixed effects in model (1.1), and $\boldsymbol{\alpha} = (\alpha_0, \alpha_1)$ denote the fixed effects in model (1.2). By treating the unobservable random effects $\boldsymbol{u}_i$ and a_i as "missing data", the log h-likelihood function of the the robust NLME model (1.1)+(1.2) can be written as

$$h(\boldsymbol{u}_i, a_i, \boldsymbol{\beta}, \boldsymbol{\alpha}, D, \sigma_a) = \sum_{i=1}^{n} \sum_{j=i}^{n_i} \log f(y_{ij} \mid \tilde{x}^*_{ij}, \boldsymbol{\beta}, \boldsymbol{\alpha}, \boldsymbol{u}_i, a_i) + \sum_{i=1}^{n} f(\boldsymbol{u}_i|D) + \sum_{i=1}^{n} f(a_i|\sigma_a), \tag{2.1}$$

where

- $f(y_{ij} \mid \tilde{x}^*_{ij}, \boldsymbol{\beta}, \boldsymbol{\alpha}, \boldsymbol{u}_i, a_i)$ is the density function of the normal distribution $\mathcal{N}(P_{i1} + P_{i2}e^{-\lambda_i t_{ij}}, \sigma^2_{ij})$. The variance σ^2_{ij} is modeled by (1.2), that is, σ^2_{ij} is assumed to be dependent on the time-varying covariate x^*_{ij} and an individual-specific effect a_i. It should be noted that, x^*_{ij}'s are replaced by their estimated values $\tilde{x}^*_{ij}$'s as if they were observed data.

- $f(\boldsymbol{u}_i|D)$ is the density of multivariate normal distribution $\mathcal{N}(\mathbf{0}, D)$ and $f(a_i|\sigma_a)$ is the density of the normal distribution $\mathcal{N}(0, \sigma^2_a)$. Here the random effects $\boldsymbol{u}_i$ and a_i are assumed to be independent.

For inference, we can obtain the parameter estimates by iterating the following steps with some starting values $(\boldsymbol{u}^{(0)}_i, a^{(0)}_i, \beta^{(0)}, \alpha^{(0)}, D^{(0)}, \sigma^{(0)}_a)$ until convergence: at iteration k,

1. solve $\partial h(\boldsymbol{u}_i, a_i, \boldsymbol{\beta}^{(k)}, \boldsymbol{\alpha}^{(k)}, D^{(k)}, \sigma^{(k)}_a)/\partial(\boldsymbol{u}_i, a_i) = 0$ to obtain the updated estimate $(\boldsymbol{u}^{(k+1)}_i, a^{(k+1)}_i)$,

2. solve $\partial h(\boldsymbol{u}^{(k+1)}_i, a^{(k+1)}_i, \boldsymbol{\beta}, \boldsymbol{\alpha}, D^{(k)}, \sigma^{(k)}_a)/\partial(\boldsymbol{\beta}, \boldsymbol{\alpha}) = 0$ to obtain the updated estimate $(\boldsymbol{\beta}^{(k+1)}, \boldsymbol{\alpha}^{(k+1)})$,

3. solve $\partial h(\boldsymbol{u}^{(k+1)}_i, a^{(k+1)}_i, \boldsymbol{\beta}^{(k+1)}, \boldsymbol{\alpha}^{(k+1)}, D, \sigma_a)/\partial(D, \sigma_a) = 0$ to obtain the updated estimate $D^{(k+1)}$.

This method may lead to two major problems: (i) the longitudinal processes for viral load and CD4 cell count may be associated with each other but the two-step inference method fits models to each process separately, so the estimates may be biased, especially when there is a strong association between the two processes; (ii) the Two-step inference method treats estimated CD4 as observed data in the second step, so the uncertainty of estimation in the first step is not incorporated and hence the SE's of the parameters may be under-estimated.

To adjust the under-estimated SE's from the Two-step inference method, we may consider a parametric bootstrap approach to get a large bootstrap sample of estimates for all parameters, and use the empirical sample SE's of the bootstrap estimates as the adjusted SE's for the parameter estimators.

Joint likelihood method

Alternatively, we can make joint inference to the robust NLME model and covariate process LME model simultaneously based on the **joint likelihood** function of the models (1.1), (1.2), and (1.3). The joint likelihood of the three models can be complicated and intractable, so we will use the h-likelihood method to approximate.

Let $\boldsymbol{v}_i = (\boldsymbol{u}_i, a_i, b_i)^T$ denote the collection all random effects in (1.1), (1.2), and (1.3), $\boldsymbol{\theta} = (\boldsymbol{\beta}, \boldsymbol{\alpha}, \boldsymbol{\gamma})$ denote the collection of all fixed mean parameters, and $\Sigma = (D, \sigma_a, \sigma_b, \xi)$ denote the collection of all fixed

variance-covariance parameters. The joint log h-likelihood of models (1.1), (1.2), and (1.3) is written as

$$h(\boldsymbol{v}_i, \boldsymbol{\theta}, \Sigma) = \sum_{i=1}^{n} \sum_{j=i}^{n_i} \log f(y_{ij} \mid x_{ij}^*, \boldsymbol{\theta}, \boldsymbol{v}_i) + \sum_{i=1}^{n} \sum_{j=i}^{n_i} \log f(x_{ij} \mid \boldsymbol{\theta}, \boldsymbol{v}_i, \Sigma) + \sum_{i=1}^{n} f(\boldsymbol{v}_i|\Sigma), \tag{2.2}$$

where

- $f(y_{ij} \mid x_{ij}^*, \boldsymbol{\theta}, \boldsymbol{v}_i)$ is the density function of the normal distribution $\mathcal{N}(P_{i1} + P_{i2}e^{-\lambda_i t_{ij}}, \sigma_{ij}^2)$. The variance σ_{ij}^2 is modeled by (1.2), that is, σ_{ij}^2 is assumed to be dependent on the time-varying covariate x_{ij}^* and an individual-specific effect a_i. It should be noted that, x_{ij}^* is defined by model (1.3);
- $f(x_{ij} \mid \boldsymbol{\theta}, \boldsymbol{v}_i, \Sigma)$ is the density function of the normal distribution $\mathcal{N}(\gamma_0 + \gamma_1 t_{ij} + \gamma_2 t_{ij}^2 + b_i, \xi^2)$;
- $f(\boldsymbol{v}_i|\Sigma)$ is the density of all random effects, which is basically a multivariate normal distribution with mean $\mathbf{0}$ and a variance-covariance matrix. The diagonal blocks of the variance-covariance matrix are $(D, \sigma_a^2, \sigma_b^2)$, and the off-diagonal elements are flexible to the assumptions to the correlation among random effects.

Likewise the Two-step inference, we can obtain the parameter estimates by iterating the following steps with some starting values $(\boldsymbol{v}_i^{(0)}, \boldsymbol{\theta}^{(0)}, \Sigma^{(0)})$ until convergence: at iteration k,

1. solve $\partial h(\boldsymbol{v}_i, \boldsymbol{\theta}^{(k)}, \Sigma^{(k)})/\partial \boldsymbol{v}_i = 0$ to obtain the updated estimate $\boldsymbol{v}_i^{(k+1)}$,
2. solve $\partial h(\boldsymbol{v}_i^{(k+1)}, \boldsymbol{\theta}, \Sigma^{(k)})/\partial \boldsymbol{\theta} = 0$ to obtain the updated estimate $\boldsymbol{\theta}^{(k+1)}$,
3. solve $\partial h(\boldsymbol{v}_i^{(k+1)}, \boldsymbol{\theta}^{(k+1)}, \Sigma)/\partial \Sigma = 0$ to obtain the updated estimate $\Sigma^{(k+1)}$.

3 Empirical Results

Table 3.1 shows the Two-step and joint likelihood inference results for models (1.1), (1.2) and (1.3) to a real dataset.

Table 3.1: Parameters estimates

Assumption for random effects	Two-step method u_{i1}, u_{i2}, u_{i3} correlated, independent from a_i				Joint likelihood method u_{i1}, u_{i2}, u_{i3} correlated, independent from a_i, b_i			
Parameters	Est.	SE[1]	SE[2]	SE[3]	Est.	SE[4]	SE[5]	SE[6]
Fixed effects in robust NLME:								
P_1	2.479	0.070	0.078	0.080	2.488	0.068	0.067	0.063
P_2	3.032	0.052	0.038	0.038	3.000	0.052	0.049	0.049
λ	7.592	0.299	0.257	0.259	7.514	0.315	0.288	0.284
Fixed effects in robust NLME variance component:								
α_0	-25.049	1.421	4.400	1.797	-17.638	1.297	2.942	1.102
α_1	4.071	0.257	0.803	0.327	2.715	0.235	0.535	0.199
Fixed effects in LME:								
γ_0	5.209	0.064	0.062	0.063	5.205	0.064	0.065	0.064
γ_1	1.574	0.203	0.228	0.231	1.554	0.172	0.178	0.178
γ_2	-1.206	0.224	0.240	0.248	-1.134	0.189	0.201	0.203
All variance-covariance parameters:								
d_1	0.360	0.037	0.056	0.055	0.361	0.036	0.055	0.049
d_2	0.122	0.044	0.040	0.035	0.099	0.047	0.044	0.041
d_3	0.760	0.216	0.228	0.192	0.867	0.233	0.266	0.257
σ_a	1.343	0.095	0.217	0.229	1.008	0.098	0.183	0.186
σ_b	0.365		0.048	0.046	0.374	0.013	0.042	0.041
ξ	0.220		0.011	0.011	0.200	0.009	0.009	0.009

[1] SE's for parameters in the robust NLME model (1.1)+(1.2) are based on the h-likelihood method, and SE's for parameters in the LME model (1.3) are from the `nlme` package.
[2] SE based on the parametric bootstrapping method. The SE is the empirical SE among estimates from 200 bootstrapping runs.
[3] SE based on the parametric bootstrapping method. Simulation runs with rBias>15% are removed. The SE is the empirical SE among estimates from 101 bootstrapping runs.
[4] SE's for parameters in model (1.1), (1.2) and (1.3) are based on the joint h-likelihood inference.
[5] SE based on the parametric bootstrapping method. The SE is the empirical SE among estimates from 200 bootstrapping runs.
[6] SE based on the parametric bootstrapping method. Simulation runs with rBias>15% are removed. The SE is the empirical SE among estimates from 115 bootstrapping runs.

4 Simulations

Simulation was conducted to compare the performance of the two above-mentioned inference methods: Two-step method and Joint model. Data are simulated for $n = 100$ subjects, with $n_i = 15$ repeated measurements for each subject. The entire study period is set to be $[0, 1]$, and the 15 measuring time points are equally allocated within the time interval. The true parameter values are chosen based on the estimates from the empirical analysis in Section 3. Specifically,

$$\begin{aligned}
\boldsymbol{\beta} &= (P_1, P_2, \lambda)^T = (2.5, 3.0, 7.5)^T, \\
\boldsymbol{\alpha} &= (\alpha_0, \alpha_1)^T = (-8, 1.5)^T, \\
\boldsymbol{\gamma} &= (\gamma_0, \gamma_1, \gamma_2)^T = (5.2, 1.6, -1.2)^T, \\
(d_1, d_2, d_3) &= (0.4, 0.2, 1.0), \quad \sigma_b = 0.4, \quad \xi = 0.2.
\end{aligned}$$

The structure of the variance-covariance matrix for the random effects $\boldsymbol{v}_i = (\boldsymbol{u}_i, a_i, b_i)$ varies depending on different assumptions. The starting values for the inference are set to be the true values. The number of simulation repetition is set to be 200.

We report "Est", "rBias", "rMSE", "SE.em", "SE", and "Coverage" of the parameters in models (1.1), (1.2) and (1.3). For each parameter, "Est" is the average of the 200 estimates from the 200 simulation runs, "rBias"

and "rMSE" are the relative bias and mean square error of the 200 parameter estimates, which are defined as follows:

$$\text{rBias (\%) of } \hat{\beta} = \left| \left(\frac{1}{M} \sum_{m=1}^{M} \hat{\beta}^{(m)} - \beta \right) / \beta \right| \times 100\%,$$

$$\text{rMSE (\%) of } \hat{\beta} = \left| \frac{1}{M} \sum_{m=1}^{M} \left(\hat{\beta}^{(m)} - \beta \right)^2 / \beta \right| \times 100\%,$$

where $\hat{\beta}^{(m)}$ is the estimate of β in the m-th simulation run and M is the total number of simulation repetitions, "SE.em" is the empirical standard deviation of the 200 estimates, "SE" is the square root of the average of the 200 variance estimates, and "Coverage" is the empirical coverage of the true parameter value among the 200 95% confidence intervals from the 200 simulation runs.

Setting 1

In this setting, we assume the random effects u_{i1}, u_{i2}, u_{i3}, and b_i are correlated by setting the correlation coefficients to be 0.5, and they are independent from a_i. We also assume $a_i \sim \mathcal{N}(0, 1)$. Table 4.1 shows the results based on all the 200 simulation runs and Table 4.2 shows the results based on the simulation runs with abnormally large bias (rBias>15%) removed.

Table 4.1: Simulation results: all 200 runs

	Two-step method						Joint likelihood method					
Parameters	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage
Fixed effects in robust NLME:												
$P_1 = 2.5$	2.481	0.762	0.161	0.061	0.047	0.855	2.517	0.664	0.146	0.058	0.046	0.890
$P_2 = 3.0$	2.978	0.724	0.254	0.085	0.066	0.840	3.032	1.082	0.235	0.078	0.068	0.880
$\lambda = 7.5$	7.447	0.711	2.891	0.464	0.393	0.885	7.616	1.544	2.672	0.433	0.388	0.910
Fixed effects in robust NLME variance component:												
$\alpha_0 = -8$	-8.584	7.303	9.113	0.624	0.495	0.740	-8.420	5.255	6.542	0.590	0.490	0.830
$\alpha_1 = 1.5$	1.583	5.537	1.311	0.113	0.087	0.770	1.554	3.609	0.944	0.106	0.086	0.855
Fixed effects in LME:												
$\gamma_0 = 5.2$	5.200	0.006	0.038	0.045	0.042	0.925	5.198	0.037	0.038	0.045	0.042	0.930
$\gamma_1 = 1.6$	1.598	0.099	0.248	0.063	0.063	0.945	1.604	0.261	0.252	0.064	0.060	0.935
$\gamma_2 = -1.2$	-1.200	0.006	0.313	0.061	0.061	0.955	-1.203	0.291	0.318	0.062	0.058	0.935

Table 4.2: Simulation results: runs with rBias>15% removed

	Two-step method[1]						Joint likelihood method[2]					
Parameters	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage
Fixed effects in robust NLME:												
$P_1 = 2.5$	2.481	0.748	0.158	0.060	0.047	0.863	2.519	0.748	0.150	0.059	0.046	0.883
$P_2 = 3.0$	2.981	0.629	0.233	0.082	0.066	0.875	3.032	1.054	0.216	0.074	0.068	0.889
$\lambda = 7.5$	7.452	0.646	2.720	0.450	0.394	0.887	7.601	1.345	2.470	0.420	0.388	0.917
Fixed effects in robust NLME variance component:												
$\alpha_0 = -8$	-8.445	5.562	5.201	0.468	0.496	0.875	-8.332	4.151	4.062	0.465	0.492	0.911
$\alpha_1 = 1.5$	1.558	3.835	0.695	0.085	0.087	0.905	1.538	2.518	0.560	0.084	0.086	0.939
Fixed effects in LME:												
$\gamma_0 = 5.2$	5.201	0.012	0.041	0.046	0.042	0.911	5.199	0.017	0.038	0.045	0.042	0.933
$\gamma_1 = 1.6$	1.599	0.075	0.242	0.062	0.063	0.952	1.602	0.153	0.248	0.063	0.060	0.933
$\gamma_2 = -1.2$	-1.201	0.082	0.301	0.060	0.061	0.970	-1.202	0.152	0.312	0.061	0.058	0.933

[1] Simulation runs with abnormally large bias (rBias>15%) are removed. 168 runs are kept for the result presented.
[2] Simulation runs with abnormally large bias (rBias>15%) are removed. 180 runs are kept for the result presented.

Setting 2

In setting 2, we assume all random effects are independent from each other. We also assume $a_i \sim \mathcal{N}(0,1)$. Table 4.3 shows the results based on all the 200 simulation runs and Table 4.4 shows the results based on the simulation runs with abnormally large bias (rBias>15%) removed.

Table 4.3: Simulation results: all 200 runs

	Two-step method						Joint likelihood method					
Parameters	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage
Fixed effects in robust NLME:												
$P_1 = 2.5$	2.515	0.601	0.152	0.060	0.046	0.875	2.515	0.599	0.153	0.060	0.046	0.865
$P_2 = 3.0$	3.006	0.207	0.261	0.089	0.068	0.870	3.006	0.200	0.269	0.090	0.068	0.860
$\lambda = 7.5$	8.109	8.126	9.761	0.602	0.364	0.580	8.113	8.174	9.575	0.587	0.363	0.575
Fixed effects in robust NLME variance component:												
$\alpha_0 = -8$	-9.061	13.261	23.484	0.870	0.500	0.490	-8.840	10.505	18.557	0.884	0.494	0.560
$\alpha_1 = 1.5$	1.665	11.005	3.344	0.152	0.087	0.550	1.625	8.361	2.603	0.153	0.086	0.615
Fixed effects in LME:												
$\gamma_0 = 5.2$	5.202	0.041	0.031	0.040	0.042	0.940	5.197	0.051	0.031	0.040	0.042	0.955
$\gamma_1 = 1.6$	1.597	0.164	0.273	0.066	0.063	0.935	1.619	1.184	0.282	0.065	0.060	0.910
$\gamma_2 = -1.2$	-1.197	0.243	0.326	0.063	0.061	0.940	-1.214	1.205	0.332	0.062	0.058	0.925

Table 4.4: Simulation results: runs with rBias>15% removed

	Two-step method[1]						Joint likelihood method[2]					
Parameters	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage
Fixed effects in robust NLME:												
$P_1 = 2.5$	2.510	0.395	0.125	0.055	0.047	0.879	2.511	0.451	0.119	0.054	0.046	0.908
$P_2 = 3.0$	2.998	0.076	0.277	0.092	0.068	0.818	3.001	0.047	0.302	0.096	0.068	0.826
$\lambda = 7.5$	7.936	5.814	4.641	0.399	0.359	0.707	7.961	6.147	5.136	0.417	0.359	0.670
Fixed effects in robust NLME variance component:												
$\alpha_0 = -8$	-8.647	8.083	7.210	0.400	0.499	0.778	-8.419	5.241	5.406	0.509	0.500	0.862
$\alpha_1 = 1.5$	1.593	6.167	0.913	0.072	0.087	0.879	1.552	3.438	0.678	0.087	0.087	0.936
Fixed effects in LME:												
$\gamma_0 = 5.2$	5.205	0.096	0.035	0.042	0.042	0.909	5.198	0.040	0.037	0.044	0.042	0.927
$\gamma_1 = 1.6$	1.597	0.174	0.311	0.071	0.063	0.909	1.620	1.245	0.352	0.073	0.060	0.862
$\gamma_2 = -1.2$	-1.196	0.360	0.379	0.068	0.061	0.909	-1.214	1.182	0.413	0.069	0.058	0.881

[1] Simulation runs with abnormally large bias (rBias>15%) are removed. Only 99 runs are kept for the result presented.
[2] Simulation runs with abnormally large bias (rBias>15%) are removed. Only 109 runs are kept for the result presented.

Setting 3

In setting 3, we assume random effects u_{i1}, u_{i2}, u_{i3} are correlated, and $\boldsymbol{u}_i, a_i, b_i$ are independent from each other. We also assume $a_i \sim \mathcal{N}(0,1)$. Table 4.5 shows the results based on all the 200 simulation runs and Table 4.6 shows the results based on the simulation runs with abnormally large bias (rBias>15%) removed.

Table 4.5: Simulation results: all 200 runs

	Two-step method						Joint likelihood method					
Parameters	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage
Fixed effects in robust NLME:												
$P_1 = 2.5$	2.499	0.039	0.143	0.060	0.048	0.875	2.502	0.089	0.139	0.059	0.047	0.870
$P_2 = 3.0$	2.991	0.293	0.271	0.090	0.066	0.865	2.992	0.252	0.275	0.091	0.067	0.865
$\lambda = 7.5$	7.539	0.525	3.381	0.503	0.392	0.860	7.663	2.168	4.089	0.531	0.399	0.845
Fixed effects in robust NLME variance component:												
$\alpha_0 = -8$	-8.648	8.096	15.423	0.905	0.497	0.595	-8.270	3.380	9.922	0.851	0.492	0.695
$\alpha_1 = 1.5$	1.596	6.373	2.316	0.160	0.087	0.630	1.528	1.858	1.569	0.151	0.086	0.725
Fixed effects in LME:												
$\gamma_0 = 5.2$	5.198	0.038	0.032	0.041	0.042	0.965	5.196	0.073	0.032	0.041	0.042	0.965
$\gamma_1 = 1.6$	1.612	0.734	0.246	0.062	0.063	0.955	1.619	1.189	0.261	0.062	0.060	0.950
$\gamma_2 = -1.2$	-1.211	0.907	0.305	0.060	0.061	0.935	-1.216	1.342	0.314	0.059	0.058	0.935

Table 4.6: Simulation results: runs with rBias>15% removed

	Two-step method[1]						Joint likelihood method[2]					
Parameters	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage
Fixed effects in robust NLME:												
$P_1 = 2.5$	2.501	0.026	0.142	0.060	0.048	0.880	2.502	0.071	0.133	0.058	0.047	0.879
$P_2 = 3.0$	2.989	0.360	0.274	0.090	0.066	0.866	2.995	0.173	0.280	0.092	0.067	0.854
$\lambda = 7.5$	7.565	0.862	3.490	0.509	0.397	0.852	7.661	2.143	4.245	0.543	0.398	0.841
Fixed effects in robust NLME variance component:												
$\alpha_0 = -8$	-8.353	4.408	5.507	0.564	0.495	0.824	-8.211	2.635	4.416	0.558	0.494	0.885
$\alpha_1 = 1.5$	1.543	2.867	0.782	0.100	0.086	0.887	1.517	1.107	0.664	0.099	0.086	0.924
Fixed effects in LME:												
$\gamma_0 = 5.2$	5.201	0.016	0.032	0.041	0.042	0.972	5.197	0.055	0.030	0.039	0.042	0.968
$\gamma_1 = 1.6$	1.612	0.732	0.273	0.065	0.063	0.951	1.617	1.074	0.267	0.063	0.060	0.943
$\gamma_2 = -1.2$	-1.210	0.870	0.347	0.064	0.061	0.923	-1.214	1.156	0.323	0.061	0.058	0.930

[1] Simulation runs with abnormally large bias (rBias>15%) are removed. 142 runs are kept for the result presented.
[2] Simulation runs with abnormally large bias (rBias>15%) are removed. 157 runs are kept for the result presented.

Setting 4

In setting 4, we assume random effects u_{i1}, u_{i2}, u_{i3} are correlated, and $\boldsymbol{u}_i, a_i, b_i$ are independent from each other. Table 4.7 shows the results based on all the 200 simulation runs and Table 4.8 shows the results based on the simulation runs with abnormally large bias (rBias>15%) removed.

Table 4.7: Simulation results: all 200 runs

	Two-step method						Joint likelihood method					
Parameters	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage
Fixed effects in robust NLME:												
$P_1 = 2.5$	2.502	0.071	0.175	0.066	0.051	0.860	2.506	0.238	0.176	0.066	0.050	0.870
$P_2 = 3.0$	2.990	0.336	0.291	0.093	0.070	0.880	2.991	0.308	0.290	0.093	0.072	0.885
$\lambda = 7.5$	7.478	0.295	3.695	0.527	0.409	0.860	7.627	1.688	4.649	0.578	0.421	0.825
Fixed effects in robust NLME variance component:												
$\alpha_0 = -8$	-9.112	13.904	27.875	0.999	0.491	0.415	-8.667	8.338	15.220	0.881	0.486	0.590
$\alpha_1 = 1.5$	1.676	11.705	4.151	0.178	0.086	0.490	1.596	6.383	2.224	0.156	0.086	0.645
Fixed effects in LME:												
$\gamma_0 = 5.2$	5.199	0.019	0.040	0.045	0.042	0.925	5.197	0.062	0.040	0.045	0.042	0.930
$\gamma_1 = 1.6$	1.603	0.178	0.265	0.065	0.063	0.940	1.611	0.705	0.272	0.065	0.060	0.935
$\gamma_2 = -1.2$	-1.202	0.190	0.320	0.062	0.061	0.955	-1.208	0.675	0.327	0.062	0.058	0.940

Table 4.8: Simulation results: runs with rBias>15% removed

	Two-step method[1]						Joint likelihood method[2]					
Parameters	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage
Fixed effects in robust NLME:												
$P_1 = 2.5$	2.505	0.199	0.203	0.071	0.051	0.843	2.509	0.355	0.160	0.063	0.050	0.890
$P_2 = 3.0$	2.988	0.395	0.293	0.093	0.071	0.889	2.992	0.282	0.279	0.091	0.073	0.897
$\lambda = 7.5$	7.478	0.297	4.015	0.551	0.413	0.861	7.668	2.242	4.519	0.559	0.427	0.828
Fixed effects in robust NLME variance component:												
$\alpha_0 = -8$	-8.495	6.186	6.478	0.525	0.484	0.769	-8.390	4.874	5.640	0.549	0.490	0.814
$\alpha_1 = 1.5$	1.565	4.348	0.841	0.092	0.085	0.907	1.547	3.109	0.783	0.098	0.086	0.890
Fixed effects in LME:												
$\gamma_0 = 5.2$	5.195	0.091	0.045	0.048	0.043	0.907	5.197	0.055	0.042	0.047	0.042	0.917
$\gamma_1 = 1.6$	1.612	0.781	0.272	0.065	0.063	0.926	1.614	0.887	0.288	0.067	0.060	0.924
$\gamma_2 = -1.2$	-1.212	0.964	0.331	0.062	0.061	0.935	-1.212	0.975	0.346	0.064	0.058	0.931

[1] Simulation runs with abnormally large bias (rBias>15%) are removed. 108 runs are kept for the result presented.
[2] Simulation runs with abnormally large bias (rBias>15%) are removed. 145 runs are kept for the result presented.

Setting 5 [parametric bootstrap, 3 by 3 covariance matrix]

In setting 5, we assume random effects u_{i1}, u_{i2}, u_{i3} are correlated, and $\boldsymbol{u}_i, a_i, b_i$ are independent from each other. In addition, **parametric** bootstrapping method was used to adjust the SE for the parameter estimates. We conducted 50 bootstrapping runs for each of 50 simulation runs. Table 4.9 shows the results based on all the 50 simulation runs and Table 4.10 shows the results based on the simulation runs with abnormally large bias (rBias>15%) removed.

In Table 4.9 and 4.10, "SE.BT" is the average of bootstrapping SE from the 50 simulation runs and " Coverage.bt" is the corresponding coverage rate. The bootstrapping SE was calculated as the empirical SE of the estimates from the 50 bootstrapping runs. "SE.BT1" ("Coverage.bt1") and "SE.BT2" ("Coverage.bt2") are defined similarly except that the bootstrapping SE for each simulation run was based on the bootstrapping runs with abnormally large bias (rBias > 10% and rBias > 15%, respectively) removed.

Table 4.9: Simulation results: all 50 runs

	Two-step method						Joint likelihood method											
Parameters	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	SE.BT	Coverage.bt	SE.BT1[a]	Coverage.bt1	SE.BT2[b]	Coverage.bt2
Fixed effects in robust NLME:																		
$P_1 = 2.5$	2.505	0.188	0.195	0.070	0.051	0.820	2.508	0.314	0.195	0.070	0.050	0.820	0.055	0.860	0.053	0.860	0.055	0.880
$P_2 = 3.0$	2.988	0.414	0.226	0.082	0.070	0.920	2.989	0.363	0.239	0.085	0.072	0.920	0.079	0.940	0.081	0.920	0.080	0.920
$\lambda = 7.5$	7.440	0.804	3.652	0.525	0.412	0.880	7.550	0.668	3.590	0.522	0.419	0.880	0.478	0.940	0.371	0.820	0.433	0.880
Fixed effects in robust NLME variance component:																		
$\alpha_0 = -8$	-9.058	13.231	26.420	1.007	0.492	0.400	-8.535	6.691	15.344	0.980	0.487	0.640	0.780	0.840	0.403	0.560	0.524	0.700
$\alpha_1 = 1.5$	1.668	11.215	4.022	0.181	0.087	0.380	1.574	4.959	2.402	0.176	0.086	0.720	0.138	0.840	0.072	0.680	0.093	0.780
Fixed effects in LME:																		
$\gamma_0 = 5.2$	5.199	0.015	0.037	0.044	0.042	0.940	5.197	0.058	0.038	0.045	0.042	0.920	0.043	0.920	0.043	0.920	0.043	0.940
$\gamma_1 = 1.6$	1.600	0.019	0.231	0.061	0.064	0.940	1.611	0.690	0.238	0.061	0.060	0.920	0.060	0.920	0.054	0.860	0.060	0.900
$\gamma_2 = -1.2$	-1.194	0.482	0.314	0.062	0.062	0.960	-1.203	0.278	0.301	0.061	0.058	0.900	0.057	0.900	0.051	0.860	0.057	0.900

[a] Bootstrapping runs with abnormally large bias (rBias>10%) are removed. The average bootstrapping runs is 24 for each simulation run.
[b] Bootstrapping runs with abnormally large bias (rBias>15%) are removed. The average bootstrapping runs is 37 for each simulation run.

Table 4.10: Simulation results: runs with rBias>15% removed

	Two-step method[1]						Joint likelihood method[2]											
Parameters	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	SE.BT	Coverage.bt	SE.BT1	Coverage.bt1	SE.BT2	Coverage.bt2
Fixed effects in robust NLME:																		
$P_1 = 2.5$	2.506	0.237	0.284	0.086	0.050	0.667	2.503	0.130	0.217	0.075	0.049	0.769	0.054	0.821	0.052	0.821	0.054	0.846
$P_2 = 3.0$	2.978	0.721	0.238	0.083	0.071	0.917	2.984	0.544	0.256	0.087	0.072	0.923	0.079	0.923	0.082	0.923	0.081	0.923
$\lambda = 7.5$	7.416	1.122	2.448	0.429	0.414	0.958	7.547	0.633	3.374	0.507	0.419	0.872	0.484	0.949	0.370	0.795	0.435	0.872
Fixed effects in robust NLME variance component:																		
$\alpha_0 = -8$	-8.444	5.553	6.418	0.574	0.488	0.792	-8.466	5.830	6.190	0.534	0.486	0.795	0.768	1.000	0.400	0.692	0.520	0.872
$\alpha_1 = 1.5$	1.561	4.091	0.990	0.108	0.086	0.750	1.563	4.192	0.854	0.095	0.086	0.897	0.136	1.000	0.072	0.846	0.093	0.974
Fixed effects in LME:																		
$\gamma_0 = 5.2$	5.191	0.171	0.040	0.046	0.042	0.958	5.197	0.058	0.038	0.045	0.042	0.923	0.043	0.897	0.044	0.923	0.043	0.923
$\gamma_1 = 1.6$	1.597	0.180	0.238	0.063	0.064	0.958	1.608	0.469	0.191	0.055	0.060	0.949	0.060	0.923	0.054	0.897	0.059	0.897
$\gamma_2 = -1.2$	-1.188	0.960	0.317	0.062	0.062	0.958	-1.199	0.082	0.250	0.055	0.058	0.949	0.057	0.923	0.051	0.897	0.057	0.923

[1] Simulation runs with abnormally large bias (rBias>15%) are removed. Only 24 runs are kept for the result presented.
[2] Simulation runs with abnormally large bias (rBias>15%) are removed. Only 39 runs are kept for the result presented.

Setting 6 [parametric bootstrap, 2 by 2 covariance matrix]

In setting 6, we drop one random effect u_{i2} from the NLME model (1.1). In other words, we assume there are two random effects u_{i1}, u_{i3} in the the NLME model, which are correlated. We also assume $\boldsymbol{u}_i = (u_{i1}, u_{i3}), a_i, b_i$ are independent from each other. In addition, **parametric** bootstrapping method was used to adjust the SE for the parameter estimates. We conducted 100 bootstrapping runs for each of 100 simulation runs. Table 4.11 shows the results based on all the 100 simulation runs and Table 4.12 shows the results based on the simulation runs with abnormally large bias (rBias>15%) removed.

"SE.BT" is the average of bootstrapping SE from the 100 simulation runs and " Coverage.bt" is the corresponding coverage rate. The bootstrapping SE was calculated as the empirical SE of the estimates from the 100 bootstrapping runs. "SE.BT1" ("Coverage.bt1") and "SE.BT2" ("Coverage.bt2") are defined similarly except that the bootstrapping SE for each simulation run was based on the bootstrapping runs with abnormally large bias (rBias > 10% and rBias > 15%, respectively) removed.

Table 4.11: Simulation results: all 100 runs

	Two-step method						Joint likelihood method											
Parameters	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	SE.BT	Coverage.bt	SE.BT1[a]	Coverage.bt1	SE.BT2[b]	Coverage.bt2
Fixed effects in robust NLME:																		
$P_1 = 2.5$	2.501	0.044	0.122	0.055	0.053	0.930	2.503	0.121	0.126	0.056	0.052	0.920	0.058	0.950	0.057	0.950	0.057	0.950
$P_2 = 3.0$	2.992	0.252	0.184	0.074	0.069	0.910	2.993	0.224	0.182	0.074	0.070	0.910	0.076	0.950	0.075	0.940	0.076	0.940
$\lambda = 7.5$	7.461	0.518	3.378	0.504	0.423	0.880	7.548	0.645	3.711	0.528	0.431	0.880	0.481	0.910	0.362	0.820	0.442	0.870
Fixed effects in robust NLME variance component:																		
$\alpha_0 = -8$	-8.279	3.489	7.580	0.731	0.488	0.810	-7.965	0.442	7.030	0.753	0.483	0.770	0.684	0.940	0.392	0.680	0.539	0.840
$\alpha_1 = 1.5$	1.527	1.833	1.159	0.130	0.086	0.800	1.471	1.912	1.235	0.134	0.085	0.780	0.121	0.920	0.070	0.620	0.096	0.860
Fixed effects in LME:																		
$\gamma_0 = 5.2$	5.201	0.023	0.032	0.041	0.043	0.970	5.200	0.004	0.032	0.041	0.043	0.970	0.043	0.970	0.042	0.950	0.042	0.970
$\gamma_1 = 1.6$	1.593	0.421	0.244	0.062	0.063	0.960	1.599	0.040	0.238	0.062	0.060	0.940	0.060	0.920	0.053	0.910	0.059	0.940
$\gamma_2 = -1.2$	-1.189	0.932	0.304	0.060	0.061	0.960	-1.194	0.537	0.288	0.059	0.058	0.930	0.057	0.940	0.051	0.880	0.057	0.930

[a] Bootstrapping runs with abnormally large bias (rBias>10%) are removed. The average bootstrapping runs is 61 for each simulation run.
[b] Bootstrapping runs with abnormally large bias (rBias>15%) are removed. The average bootstrapping runs is 88 for each simulation run.

Table 4.12: Simulation results: runs with rBias>15% removed

	Two-step method[1]						Joint likelihood method[2]											
Parameters	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	SE.BT	Coverage.bt	SE.BT1	Coverage.bt1	SE.BT2	Coverage.bt2
Fixed effects in robust NLME:																		
$P_1 = 2.5$	2.502	0.085	0.124	0.056	0.053	0.920	2.505	0.210	0.124	0.056	0.052	0.910	0.058	0.944	0.057	0.944	0.057	0.944
$P_2 = 3.0$	2.996	0.148	0.171	0.072	0.069	0.920	2.994	0.211	0.185	0.075	0.070	0.899	0.076	0.944	0.075	0.933	0.076	0.933
$\lambda = 7.5$	7.456	0.583	3.119	0.484	0.422	0.885	7.519	0.259	3.516	0.516	0.429	0.888	0.478	0.921	0.360	0.820	0.439	0.876
Fixed effects in robust NLME variance component:																		
$\alpha_0 = -8$	-8.148	1.851	4.553	0.588	0.485	0.908	-8.017	0.213	4.766	0.621	0.482	0.843	0.682	0.989	0.394	0.742	0.541	0.910
$\alpha_1 = 1.5$	1.504	0.271	0.728	0.105	0.085	0.897	1.480	1.307	0.820	0.110	0.085	0.854	0.121	0.989	0.070	0.674	0.096	0.921
Fixed effects in LME:																		
$\gamma_0 = 5.2$	5.200	0.001	0.033	0.042	0.043	0.966	5.201	0.023	0.031	0.040	0.043	0.966	0.043	0.966	0.042	0.955	0.042	0.966
$\gamma_1 = 1.6$	1.594	0.381	0.231	0.061	0.063	0.966	1.601	0.038	0.225	0.060	0.060	0.955	0.059	0.933	0.053	0.921	0.059	0.955
$\gamma_2 = -1.2$	-1.189	0.924	0.295	0.059	0.061	0.966	-1.195	0.438	0.275	0.058	0.058	0.944	0.057	0.944	0.051	0.888	0.056	0.944

[1] Simulation runs with abnormally large bias (rBias>15%) are removed. 87 runs are kept for the result presented.
[2] Simulation runs with abnormally large bias (rBias>15%) are removed. 89 runs are kept for the result presented.

Setting 7 [non-parametric bootstrap, 3 by 3 covariance matrix]

In setting 7, we assume random effects u_{i1}, u_{i2}, u_{i3} are correlated, and $\boldsymbol{u}_i, a_i, b_i$ are independent from each other. In addition, **non-parametric** bootstrapping method was used to adjust the SE for the parameter estimates. For each **non-parametric** bootstrapping run, we re-sampled patients with replacement from the simulated data sets and fit the model. We conducted 50 bootstrapping runs for each of 100 simulation runs. Table 4.13 shows the results based on all the 100 simulation runs and Table 4.14 shows the results based on the simulation runs with abnormally large bias (rBias>15%) removed.

In the tables, "SE.BT" is the average of bootstrapping SE from the 100 simulation runs and " Coverage.bt" is the corresponding coverage rate. The bootstrapping SE was calculated as the empirical SE of the estimates from the 50 bootstrapping runs. "SE.BT1" ("Coverage.bt1") and "SE.BT2" ("Coverage.bt2") are defined similarly except that the bootstrapping SE for each simulation run was based on the bootstrapping runs with abnormally large bias (rBias > 10% and rBias > 15%, respectively) removed.

Table 4.13: Simulation results: all 100 runs

	Two-step method						Joint likelihood method											
Parameters	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	SE.BT	Coverage.bt	SE.BT1[a]	Coverage.bt1	SE.BT2[b]	Coverage.bt2
Fixed effects in robust NLME:																		
$P_1 = 2.5$	2.510	0.381	0.153	0.061	0.051	0.930	2.515	0.598	0.158	0.061	0.050	0.890	0.064	0.940	0.058	0.890	0.061	0.900
$P_2 = 3.0$	3.013	0.446	0.210	0.079	0.070	0.900	3.013	0.439	0.210	0.079	0.072	0.900	0.092	0.970	0.090	0.970	0.091	0.980
$\lambda = 7.5$	7.588	1.177	3.520	0.509	0.415	0.860	7.760	3.467	4.678	0.535	0.427	0.810	0.582	0.950	0.378	0.790	0.481	0.880
Fixed effects in robust NLME variance component:																		
$\alpha_0 = -8$	-9.016	12.704	23.880	0.941	0.488	0.460	-8.629	7.866	13.981	0.854	0.483	0.560	0.867	0.870	0.417	0.490	0.559	0.620
$\alpha_1 = 1.5$	1.660	10.670	3.633	0.171	0.086	0.530	1.591	6.040	2.163	0.156	0.085	0.570	0.154	0.900	0.074	0.530	0.099	0.640
Fixed effects in LME:																		
$\gamma_0 = 5.2$	5.195	0.104	0.029	0.039	0.043	0.970	5.192	0.146	0.029	0.038	0.043	0.960	0.042	0.950	0.042	0.940	0.042	0.950
$\gamma_1 = 1.6$	1.596	0.245	0.306	0.070	0.063	0.940	1.606	0.383	0.294	0.069	0.060	0.930	0.062	0.930	0.055	0.900	0.060	0.930
$\gamma_2 = -1.2$	-1.197	0.208	0.393	0.069	0.061	0.930	-1.206	0.467	0.371	0.067	0.058	0.910	0.059	0.920	0.052	0.860	0.058	0.910

[a] Bootstrapping runs with abnormally large bias (rBias>10%) are removed. The average bootstrapping runs is 24 for each simulation run.
[b] Bootstrapping runs with abnormally large bias (rBias>15%) are removed. The average bootstrapping runs is 37 for each simulation run.

Table 4.14: Simulation results: runs with rBias>15% removed

	Two-step method[1]						Joint likelihood method[2]											
Parameters	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	SE.BT	Coverage.bt	SE.BT1	Coverage.bt1	SE.BT2	Coverage.bt2
Fixed effects in robust NLME:																		
$P_1 = 2.5$	2.513	0.525	0.162	0.063	0.051	0.931	2.509	0.347	0.149	0.061	0.050	0.902	0.063	0.934	0.058	0.902	0.060	0.902
$P_2 = 3.0$	3.012	0.396	0.200	0.077	0.070	0.897	3.022	0.718	0.215	0.078	0.072	0.902	0.092	0.967	0.091	0.951	0.091	0.967
$\lambda = 7.5$	7.627	1.700	3.172	0.475	0.420	0.879	7.699	2.650	3.514	0.477	0.423	0.869	0.558	0.984	0.372	0.852	0.468	0.934
Fixed effects in robust NLME variance component:																		
$\alpha_0 = -8$	-8.461	5.759	6.447	0.556	0.479	0.776	-8.318	3.978	5.296	0.572	0.488	0.869	0.872	1.000	0.402	0.754	0.549	0.918
$\alpha_1 = 1.5$	1.561	4.074	0.937	0.102	0.084	0.897	1.535	2.334	0.813	0.106	0.086	0.885	0.155	1.000	0.071	0.787	0.097	0.902
Fixed effects in LME:																		
$\gamma_0 = 5.2$	5.192	0.145	0.027	0.037	0.043	0.966	5.190	0.197	0.028	0.037	0.042	0.967	0.041	0.951	0.041	0.951	0.041	0.951
$\gamma_1 = 1.6$	1.603	0.195	0.253	0.064	0.063	0.948	1.605	0.318	0.289	0.068	0.060	0.934	0.061	0.918	0.055	0.869	0.060	0.918
$\gamma_2 = -1.2$	-1.201	0.105	0.343	0.065	0.061	0.948	-1.203	0.226	0.356	0.066	0.058	0.918	0.059	0.934	0.052	0.869	0.057	0.918

[1] Simulation runs with abnormally large bias (rBias>15%) are removed. Only 48 runs are kept for the result presented.
[2] Simulation runs with abnormally large bias (rBias>15%) are removed. Only 61 runs are kept for the result presented.

Setting 8 [parametric bootstrap, 2 by 2 covariance matrix]

The setting 8 is the same as setting 6, in which we drop one random effect u_{i2} from the NLME model (1.1). In other words, we assume there are two random effects u_{i1}, u_{i3} in the the NLME model, which are correlated. We also assume $\boldsymbol{u}_i = (u_{i1}, u_{i3}), a_i, b_i$ are independent from each other. In addition, **parametric** bootstrapping method was used to adjust the SE for the parameter estimates. We conducted 100 bootstrapping runs for each of 100 simulation runs. Table 4.15 shows the results based on all the 100 simulation runs and Table 4.16 shows the results based on the simulation runs with abnormally large bias (rBias>15%) removed.

"SE.BT" is the average of bootstrapping SE from the 100 simulation runs and " Coverage.bt" is the corresponding coverage rate. "SE.BT2" ("Coverage.bt2") is defined similarly except that the bootstrapping SE for each simulation run was based on the bootstrapping runs with abnormally large bias (rBias > 15%) removed. The results from the `nlme` package are also included for comparison.

Table 4.15: Simulation results: all 100 runs

	`nlme` package						Two-step method						Joint likelihood method							
Parameters	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	SE.BT	Coverage.bt
Fixed effects in robust NLME:																				
$p_1 = 2.5$	2.527	1.076	0.235	0.072	0.067	0.920	2.509	0.375	0.102	0.050	0.052	0.960	2.511	0.428	0.106	0.051	0.052	0.950	0.057	0.970
$p_2 = 3.0$	2.996	0.135	0.441	0.116	0.148	0.990	2.994	0.190	0.222	0.082	0.069	0.920	2.995	0.174	0.222	0.082	0.069	0.930	0.075	0.960
$\lambda = 7.5$	8.054	7.393	14.313	0.880	0.862	0.910	7.699	2.655	3.647	0.486	0.433	0.930	7.759	3.455	4.027	0.487	0.441	0.920	0.495	0.950
Fixed effects in robust NLME variance component:																				
$\alpha_0 = -8$							-8.392	4.894	8.426	0.725	0.492	0.760	-8.069	0.866	7.479	0.774	0.486	0.800	0.690	0.930
$\alpha_1 = 1.5$							1.547	3.106	1.252	0.130	0.087	0.800	1.489	0.729	1.267	0.138	0.086	0.810	0.122	0.920
Fixed effects in LME:																				
$\gamma_0 = 5.2$							5.206	0.120	0.033	0.041	0.042	0.980	5.205	0.091	0.033	0.041	0.042	0.980	0.043	0.970
$\gamma_1 = 1.6$							1.598	0.144	0.207	0.058	0.063	0.970	1.603	0.210	0.193	0.056	0.060	0.970	0.061	0.960
$\gamma_2 = -1.2$							-1.198	0.167	0.258	0.056	0.061	0.960	-1.202	0.149	0.237	0.054	0.058	0.960	0.059	0.970

Table 4.16: Simulation results: runs with rBias>15% removed

	`nlme` package[1]						Two-step method[2]						Joint likelihood method[3]							
Parameters	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	Est.	rBias(%)	rMSE(%)	SE.em	SE	Coverage	SE.BT2[b]	Coverage.bt2
Fixed effects in robust NLME:																				
$p_1 = 2.5$	2.514	0.561	0.155	0.061	0.067	0.958	2.509	0.377	0.111	0.052	0.052	0.952	2.514	0.568	0.106	0.050	0.052	0.953	0.057	0.953
$p_2 = 3.0$	2.994	0.193	0.408	0.111	0.146	0.986	2.994	0.206	0.227	0.083	0.069	0.916	2.997	0.114	0.234	0.084	0.069	0.919	0.075	0.942
$\lambda = 7.5$	7.744	3.248	4.641	0.541	0.816	1.000	7.700	2.668	2.778	0.413	0.432	0.976	7.767	3.565	3.148	0.408	0.439	0.953	0.452	0.977
Fixed effects in robust NLME variance component:																				
$\alpha_0 = -8$							-8.337	4.207	4.475	0.498	0.488	0.892	-8.163	2.041	4.102	0.552	0.485	0.884	0.545	0.965
$\alpha_1 = 1.5$							1.537	2.463	0.628	0.090	0.086	0.940	1.506	0.400	0.654	0.099	0.085	0.895	0.096	0.942
Fixed effects in LME:																				
$\gamma_0 = 5.2$							5.207	0.144	0.034	0.042	0.043	0.976	5.206	0.107	0.035	0.043	0.043	0.977	0.042	0.965
$\gamma_1 = 1.6$							1.597	0.211	0.202	0.057	0.063	0.976	1.600	0.016	0.192	0.056	0.060	0.977	0.060	0.942
$\gamma_2 = -1.2$							-1.198	0.168	0.252	0.055	0.061	0.964	-1.199	0.114	0.230	0.053	0.058	0.977	0.058	0.977

[1] Simulation runs with abnormally large bias (rBias>15%) are removed. 72 runs are kept for the result presented
[2] Simulation runs with abnormally large bias (rBias>15%) are removed. 83 runs are kept for the result presented
[3] Simulation runs with abnormally large bias (rBias>15%) are removed. 86 runs are kept for the result presented
[b] Bootstrapping runs with abnormally large bias (rBias>15%) are removed. The average bootstrapping runs is 88 for each simulation run.